MedAssist AI

Advanced Agentic RAG for Medical Knowledge & Clinical Decision Support

Overview

MedAssist AI is a clinical knowledge and decision-support platform. Healthcare professionals ask medical questions and receive grounded, cited answers retrieved from an indexed knowledge base of trusted documents. It is a prototype and is not a diagnostic or prescribing system.

Architecture

Backend: Python 3.12+, FastAPI, SQLAlchemy (SQLite/PostgreSQL), Qdrant (vectors), BM25, Ollama (bge-m3 embeddings, phi3 LLM). Frontend: React + TypeScript + Vite + Tailwind (frontend/). Three RAG modes share one grounded generator so results stay comparable.

RAG Modes

Traditional RAG
Dense vector search only. System A baseline for benchmarking.

Hybrid RAG
BM25 + dense -> Reciprocal Rank Fusion -> dedupe -> BGE reranker (ONNX/CPU). Best precision.

Agentic RAG
LangGraph pipeline: safety input, planner (sub-queries), hybrid retrieval, verifier, correction loop (max 3), clinical evidence, safety agent, response generator.

Key Features

- Document ingestion: PDF/TXT/MD upload, validation, cleaning, heading-aware chunking, embedding, Qdrant indexing; duplicate detection, reindex, delete.
- Hybrid retrieval diagnostics: per-stage latency/candidate counts, reranker status, version filtering.
- Version-aware filtering via explicit document_key (superseded editions excluded when enabled).
- Execution traces persisted per request (steps, latencies, tokens, status).
- Health endpoint: database, Qdrant, embeddings, LLM, BM25 index, reranker.
- Frontend: Dashboard (live KPIs + charts), Medical Assistant (3 modes), Agent Workflow, Knowledge Base, Benchmark Center, Observability, Settings.

Implementation Phases

- Phase 1 - Ingestion + Traditional RAG (complete).
- Phase 2 - Hybrid retrieval + BGE reranker + version filtering (complete).
- Phase 3 - Agentic LangGraph pipeline (complete).
- Phase 4 - Benchmark (Traditional vs Agentic), LLM-as-judge metrics, results storage (complete).
- Phase 5 - Frontend (React/TS/Vite/Tailwind) (in progress).
- Phase 6 - CI/CD, observability hardening (planned).

How to Use

1. Start backend: .venv\Scripts\python.exe -m uvicorn app.main:app --port 8000
2. Start frontend: cd frontend && npm run dev (http://localhost:5173)
3. Seed docs: .venv\Scripts\python.exe scripts/seed_sample_docs.py
4. Open Medical Assistant, choose a mode, ask a clinical question.
5. Inspect retrieval only: POST /api/retrieval/search with {query, top_k}.
6. Run benchmark: POST /api/benchmark/run; poll /api/benchmark/results.

Safety

MedAssist AI provides evidence-based medical information for clinical decision support. It does not replace professional medical judgment, diagnosis, or treatment. Urgent symptoms trigger escalation warnings; insufficient evidence returns an explicit insufficient-evidence answer.

Example Questions

- What are the contraindications of metformin in renal impairment?
- Compare blood pressure targets for adults with type 2 diabetes (2019 vs 2024).
- What information is needed to assess fever, cough, and shortness of breath?
- Antimicrobial stewardship principles for respiratory infections.

API Endpoints (base /api)

- GET /health - component status
- POST /documents/upload - multipart (file + source, ...)
- GET /documents - paginated list
- DELETE /documents/{id}
- POST /documents/reindex
- POST /chat/traditional - dense RAG
- POST /chat/hybrid - BM25+dense -> RRF -> reranker
- POST /chat/agentic - LangGraph agentic pipeline
- POST /retrieval/search - retrieval only + diagnostics
- GET /agents/traces - execution traces
- POST /benchmark/run / GET /benchmark/results

Configuration Highlights

TOP_K=4, MAX_CONTEXT_CHARS=6000, LLM_MAX_TOKENS=600, LLM_TIMEOUT_SECONDS=600. RRF_K=60, DENSE_CANDIDATE_K=30, KEYWORD_CANDIDATE_K=30, RERANK_CANDIDATES=20. LATEST_VERSIONS_ONLY=false. RERANKER_PROVIDER=local (none|local|remote). EMBEDDING_PROVIDER=ollama (bge-m3, 1024-dim). LLM_PROVIDER=ollama (phi3:latest).

Troubleshooting

- First query is slow: model cold start on CPU; later queries reuse the loaded model.
- collection dimension does not match: embedding model changed; remove data/qdrant and restart.
- Qdrant lock error: another process holds the path; stop it or use remote Qdrant.
- reranker_used=false: reranker disabled/timed out/failed; answer uses fused RRF order.
- 501 not_implemented: capability belongs to a later phase.

Project Layout

app/ (api, core, db, schemas, services, utils), frontend/ (React/TS/Vite), sample_documents/, scripts/, tests/, docker-compose.yml, requirements*.txt, .env.example.